CS 363M Machine Learning

Final Project Expectations

The University of Texas at Austin

Fall 2026

Key Dates

- **November 19th, 2026** — **Jupyter Notebook & Process Writeup Due** by Midnight (Slip days cannot be used)
- **December 1st, 2026** — Q&A Session (Groups will sign up for time slots)

Project Overview

Your group will apply machine learning techniques to your chosen dataset using **classification and/or regression and/or clustering**. You will try a variety of algorithms, compare their results, and use appropriate metrics to measure model performance. Apply techniques learned in class (feature engineering, hyperparameter tuning, ensembling, etc.) to improve your results.

Important Notes

Grading Focus

You will **not** be explicitly graded on the performance of your models. If you achieve only 60% classification accuracy, that is acceptable—a low accuracy will not negatively affect your grade. Instead, you will be graded on your **methodology**: What is the process you used? What things did you do to try to improve the performance of the model? Show us your due diligence in dataset preparation, exploration, feature engineering, modeling, and analysis.

Use Established Libraries

You **must** use established machine learning libraries (such as `scikit-learn`, `TensorFlow`, `PyTorch`, etc.) to implement your algorithms. Do **not** hardcode algorithm implementations from scratch. The focus of this project is on your understanding of ML methodology and problem-solving, not on re-implementing standard algorithms.

AI Usage Policy

For the Jupyter Notebook: AI tools (such as ChatGPT, GitHub Copilot, etc.) are allowed for code generation, debugging, and explanations. You may use AI to help understand concepts or troubleshoot code. However, you must understand what your code does and be able to explain your approach and results during the Q&A session.

For the Process Writeup: AI tools are **NOT allowed**. Your process writeup must be written entirely by your team members in your own words. This ensures you are reflecting on your own methodology and decision-making process.

Final Deliverable

You will submit:

- A **Jupyter Notebook** that demonstrates your technical implementation and analysis
- A **Process Writeup** (PDF, max 5 pages) detailing your ML methodology and decisions

Jupyter Notebook Requirements

Your Jupyter notebook is the technical demonstration of your work: show your code, visualizations, and analysis results. Use markdown cells to briefly explain your code and interpret your findings. The methodology and decision-making rationale belong in your process writeup, not in the notebook.

What to Include:

- **Introduction:** Brief problem statement and dataset description.
- **Data Exploration & Preprocessing:** Show the code, data exploration outputs, and visualizations. Display what you cleaned and engineered. Include visualizations of data characteristics.
- **Modeling:** Display your model implementations, code, and results. Show how different algorithms performed. Include relevant outputs and comparisons.
- **Results & Analysis:** Show model outputs, performance metrics, visualizations of results, error analysis, and insights. Interpret what you found.

Note: Your notebook and writeup have different purposes. The notebook demonstrates your technical implementation and analysis (code, outputs, visualizations). The writeup explains your methodology and decision-making process (why you chose each approach).

Process Writeup Requirements

Your team should write a clear, detailed summary (max 5 pages, PDF format) that describes your entire machine learning process. This writeup should be written by your team members and detail:

- **Data exploration:** What did you discover about the data? What were your key observations?
- **Data cleaning and preprocessing:** What issues did you address? Why?
- **Feature engineering:** What features did you create or modify? What was your rationale?
- **Algorithm selection:** Which algorithms did you try? Why did you choose them?

- **Hyperparameter tuning:** What tuning did you perform? What parameters did you adjust and why?
- **Results and analysis:** How did your models perform? What patterns or insights did you find?
- **Work split:** Clearly indicate which team member was responsible for which parts of the project (e.g., "Alice: data exploration and cleaning; Bob: feature engineering and modeling; Carol: results analysis and writeup editing")

The writeup should be a narrative that captures your methodology and decision-making process, not simply a copy of outputs from your notebook. The work split section should appear at the end of the writeup and clearly delineate each team member's contributions.

Q&A Session

All groups will participate in a Q&A session on **December 1st**. Groups will sign up for available time slots. During the Q&A, be prepared to discuss your project, methodology, and results.

Grading Rubric

Important: Your grade is based on your **Jupyter Notebook AND your Q&A performance combined**. The Q&A session gives us an opportunity to verify that you understand your work and can explain your process, decisions, and results. This ensures the notebook reflects genuine understanding, not just copied code.

Data Preparation (40 points)

- **Data Cleaning (10 pts):** Was appropriate data obtained for the problem? Was data cleaning performed thoroughly and correctly?
- **Data Exploration (10 pts):** Was data sufficiently explored beyond just cleaning?
- **Feature Engineering (20 pts):** Were features engineered appropriately for the problem?

Note: Some datasets require much less data preparation than others. If your dataset requires minimal cleaning and/or feature engineering, earn these points through additional work such as: more thorough data exploration, anomaly detection, evaluation of various clustering algorithms, advanced modeling techniques, or other relevant enhancements.

Modeling (40 points)

Were correct machine learning techniques used? Is the code correct? Was a comparison of various models performed? Were models tuned to increase performance?

Code & Notebook Quality (10 points)

What is the quality of the code? Is the notebook well narrated and easy to follow? Are appropriate outputs, visualizations, and/or markdown cells used to communicate the process and results? Is the process writeup clear, thorough, and well-written?

Outcome/Results (10 points)

Was the project complete and thorough? How well was the data modeled to solve the problem? Are the results interpreted correctly?

Peer Evaluation (TBD)

Each team member will assess the contribution of other team members. Peer assessments will be used to adjust final grades at the instructor's discretion. Any group members who do not participate will receive a 0 for the project grade.

Recommended Resources

Feature Engineering

- The Comprehensive Guide for Feature Engineering
- Feature Engineering Part 1 — Continuous Data
- Feature Engineering Part 2 — Categorical Data

Machine Learning Resources

- Nine Rules Not to Follow: Common Pitfalls in Data Mining
- Getting Started with Scikit-Learn
- Machine Learning Performance Improvement Cheat Sheet
- Hyperparameter Tuning for Machine Learning Models
- How to Handle Class Imbalances